**3** **(a)** Fig. 3.1 shows a boat stored in a shed. The boat is suspended from the ceiling of the shed by two ropes.

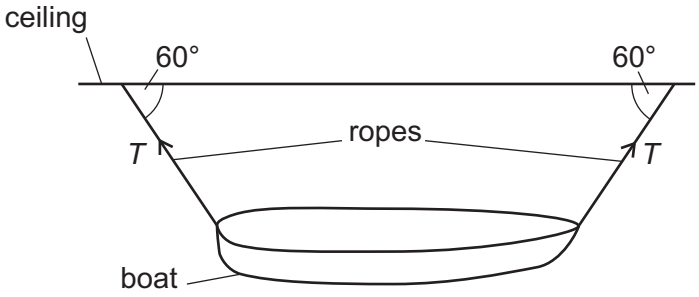



**Fig. 3.1**

The tension $T$ in each of the ropes is 75 N.

**(i)** Draw a vector diagram to determine the resultant of the forces exerted by the two ropes on the boat. State the scale you used.

scale = .....

magnitude of resultant force = .....

direction of resultant force = ..... [4]

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**(ii)** Determine the mass of the boat.

mass = ..... [1]

**(b)** Force is a vector.

Draw a circle around **two** other quantities in the list which are vectors.

**acceleration** **density** **energy** **mass**

**momentum** **power** **refractive index**

[2]

[Total: 7]